

2501.06662 — formula report

386 inline formulas, 60 display equations, 0 tables, 0 unrecovered image regions.

Display equations

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
2501.06662_E00051	021	0.536	$\begin{aligned} \operatorname{Mag}(t \mathcal{M}) &= \sum_{x, y \in \operatorname{ob}(\mathcal{M})} \zeta_t^{-1}(x, y) \\ &= \sum_{x, y \in \operatorname{ob}(\mathcal{M})} \sum_{k \geq 0} \sum_{\substack{\text{nondeg. paths} \\ x=y_0 \rightarrow y_1 \rightarrow \cdots \rightarrow y_k=y}} (-1)^k \prod_{i=1}^k \pi(y_i \mid y_{i-1})^t. \end{aligned}$	$\operatorname{Mag}(t \mathcal{M}) = \sum_{x, y \in \operatorname{ob}(\mathcal{M})} \zeta_t^{-1}(x, y)$ $= \sum_{x, y \in \operatorname{ob}(\mathcal{M})} \sum_{k \geq 0} \sum_{\substack{\text{nondeg. paths} \\ x=y_0 \rightarrow y_1 \rightarrow \cdots \rightarrow y_k=y}} (-1)^k \prod_{i=1}^k \pi(y_i \mid y_{i-1})^t.$	$\operatorname{Mag}(t \mathcal{M}) = \sum_{x, y \in \operatorname{ob}(\mathcal{M})} \zeta_t^{-1}(x, y)$ $= \sum_{x, y \in \operatorname{ob}(\mathcal{M})} \sum_{k \geq 0} \sum_{\substack{\text{nondeg. paths} \\ x=y_0 \rightarrow y_1 \rightarrow \cdots \rightarrow y_k=y}} (-1)^k \prod_{i=1}^k \pi(y_i \mid y_{i-1})^t.$
2501.06662_E00059	023	0.550	$\begin{aligned} \lim_{n \rightarrow \infty} \operatorname{Mag}(\mathcal{M}, \pi(- \mid \perp) _{T(\perp)}, \zeta) &= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{(a_1, \dots, a_{N-1}) \in A^{N-1}} \pi(\perp a_1 \cdots a_{N-1} \mid \perp) \log \pi(\perp a_1 \cdots a_{N-1} \mid \perp) \end{aligned}$	$\lim_{n \rightarrow \infty} \frac{\mathcal{H}(\mathcal{M}, \pi(- \mid \perp) _{T(\perp)}, \zeta)}{n}$ $= \lim_{n \rightarrow \infty} -\frac{1}{n} \sum_{(a_1, \dots, a_{N-1}) \in A^{N-1}} \pi(\perp a_1 \cdots a_{N-1} \mid \perp) \log \pi(\perp a_1 \cdots a_{N-1} \mid \perp)$	$\lim_{n \rightarrow \infty} \frac{\mathcal{H}(\mathcal{M}, \pi(- \mid \perp) _{T(\perp)}, \zeta)}{n}$ $= \lim_{n \rightarrow \infty} -\frac{1}{n} \sum_{(a_1, \dots, a_{N-1}) \in A^{N-1}} \pi(\perp a_1 \cdots a_{N-1} \mid \perp) \log \pi(\perp a_1 \cdots a_{N-1} \mid \perp)$
2501.06662_E00058	023	0.594	$\begin{aligned} \operatorname{Mag}(\mathcal{M}, p, \zeta) &= - \sum_{y \in \operatorname{ob}(\mathcal{M})} p(y) \log \left(\sum_{z \in \operatorname{ob}(\mathcal{M})} \zeta_t(y, z) p(z) \right) \\ &= - \sum_{y \in \operatorname{ob}(\mathcal{M})} p(y) \log \left(\sum_{z \in \operatorname{ob}(\mathcal{M})} \pi(z \mid y)^t p(z) \right). \end{aligned}$	$\mathcal{H}(\mathcal{M}, p, \zeta) = - \sum_{y \in \operatorname{ob}(\mathcal{M})} p(y) \log \left(\sum_{z \in \operatorname{ob}(\mathcal{M})} \zeta_t(y, z) p(z) \right)$ $= - \sum_{y \in \operatorname{ob}(\mathcal{M})} p(y) \log \left(\sum_{z \in \operatorname{ob}(\mathcal{M})} \pi(z \mid y)^t p(z) \right).$	$\mathcal{H}(\mathcal{M}, p, \zeta) = - \sum_{y \in \operatorname{ob}(\mathcal{M})} p(y) \log \left(\sum_{z \in \operatorname{ob}(\mathcal{M})} \zeta_t(y, z) p(z) \right)$ $= - \sum_{y \in \operatorname{ob}(\mathcal{M})} p(y) \log \left(\sum_{z \in \operatorname{ob}(\mathcal{M})} \pi(z \mid y)^t p(z) \right).$
2501.06662_E00023	015	0.636	$\begin{aligned} \# \left\{ x \in \operatorname{ob}(\mathcal{L}) : x = \perp a \text{ with } a \in \bigcup_{i=0}^{N-2} A^i \right\} (\#A + 1) &= \left(\sum_{i=0}^{N-2} \#A^i \right) (\#A + 1) \\ &= \#A^{N-1} + 2(\#A^{N-2}) + \cdots + 2(\#A) + 1. \end{aligned}$	$\# \left\{ x \in \operatorname{ob}(\mathcal{L}) : x = \perp a \text{ with } a \in \bigcup_{i=0}^{N-2} A^i \right\} (\#A + 1) = \left(\sum_{i=0}^{N-2} \#A^i \right) (\#A + 1)$ $= \#A^{N-1} + 2(\#A^{N-2}) + \cdots + 2(\#A) + 1$	$\# \left\{ x \in \operatorname{ob}(\mathcal{L}) : x = \perp a \text{ with } a \in \bigcup_{i=0}^{N-2} A^i \right\} (\#A + 1) = \left(\sum_{i=0}^{N-2} \#A^i \right) (\#A + 1)$ $= \#A^{N-1} + 2(\#A^{N-2}) + \cdots + 2(\#A) + 1.$
2501.06662_E00052 (13)	021	0.693	$\operatorname{Mag}(t \mathcal{M}) = \sum_{k \geq 0} \sum_{\ell \in [0, \infty)} \sum_{c \in G_{k, \ell}} (-1)^k e^{-t \ell}.$	$\operatorname{Mag}(t \mathcal{M}) = \sum_{k \geq 0} \sum_{\ell \in [0, \infty)} \sum_{c \in G_{k, \ell}} (-1)^k e^{-t \ell}.$	$\operatorname{Mag}(t \mathcal{M}) = \sum_{k \geq 0} \sum_{\ell \in [0, \infty)} \sum_{c \in G_{k, \ell}} (-1)^k e^{-t \ell}.$
Conf. MathPix confidence: 0.9 0.5–0.9 <0.5 — no value					

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2501.06662_E00033	017	■ 0.720	$\zeta_t^{-1}(x, y) = \begin{cases} -\pi(y \mid x)^t & \text{if } y \in \mathcal{L}_x^{(1)} \\ 1 & \text{if } y = x \\ 0 & \text{otherwise} \end{cases}$	$\zeta_t^{-1}(x, y) = \begin{cases} -\pi(y \mid x)^t & \text{if } y \in \mathcal{L}_x^{(1)} \\ 1 & \text{if } y = x \\ 0 & \text{otherwise} \end{cases}$	$\zeta_t^{-1}(x, y) = \begin{cases} -\pi(y \mid x)^t & \text{if } y \in \mathcal{L}_x^{(1)} \\ 1 & \text{if } y = x \\ 0 & \text{otherwise.} \end{cases}$
2501.06662_E00037 (10)	018	■ 0.780	$\begin{aligned} \operatorname{Mag}(t \mathcal{M}) &= \sum_{x, y \in \operatorname{ob}(\mathcal{M})} \zeta_t^{-1}(x, y) \\ &= \sum_{x \in \operatorname{ob}(\mathcal{M})} \zeta_t^{-1}(x, x) - \sum_{x \in \operatorname{ob}(\mathcal{M}) \setminus T(\perp)} \sum_{y \in \mathcal{L}_x^{(1)}} \pi(y \mid x)^t \\ &= \sum_{x \in \operatorname{ob}(\mathcal{M}) \setminus T(\perp)} \left(1 - \sum_{a \in A \cup \{\dagger\}} p_x(a)^t \right) + \#(T(\perp)). \end{aligned}$	$\begin{aligned} \operatorname{Mag}(t \mathcal{M}) &= \sum_{x, y \in \operatorname{ob}(\mathcal{M})} \zeta_t^{-1}(x, y) \\ &= \sum_{x \in \operatorname{ob}(\mathcal{M})} \zeta_t^{-1}(x, x) - \sum_{x \in \operatorname{ob}(\mathcal{M}) \setminus T(\perp)} \sum_{y \in \mathcal{L}_x^{(1)}} \pi(y \mid x)^t \\ &= \sum_{x \in \operatorname{ob}(\mathcal{M}) \setminus T(\perp)} \left(1 - \sum_{a \in A \cup \{\dagger\}} p_x(a)^t \right) + \#(T(\perp)). \end{aligned}$	$\begin{aligned} \operatorname{Mag}(t \mathcal{M}) &= \sum_{x, y \in \operatorname{ob}(\mathcal{M})} \zeta_t^{-1}(x, y) \\ &= \sum_{x \in \operatorname{ob}(\mathcal{M})} \zeta_t^{-1}(x, x) - \sum_{x \in \operatorname{ob}(\mathcal{M}) \setminus T(\perp)} \sum_{y \in \mathcal{L}_x^{(1)}} \pi(y \mid x)^t \\ &= \sum_{x \in \operatorname{ob}(\mathcal{M}) \setminus T(\perp)} \left(1 - \sum_{a \in A \cup \{\dagger\}} p_x(a)^t \right) + \#(T(\perp)). \end{aligned}$
2501.06662_E00049	020	■ 0.797	$\begin{aligned} \partial_k^i(y_0, \dots, y_k) &= \begin{cases} (y_0, \dots, y_{i-1}, y_{i+1}, \dots, y_k) & \text{if } d(y_{i-1}, y_i) + d(y_i, y_{i+1}) = d(y_{i-1}, y_{i+1}) \\ 0 & \text{otherwise} \end{cases} \end{aligned}$	$\partial_k^i(y_0, \dots, y_k) = \begin{cases} (y_0, \dots, y_{i-1}, y_{i+1}, \dots, y_k) & \text{if } d(y_{i-1}, y_i) + d(y_i, y_{i+1}) = d(y_{i-1}, y_{i+1}) \\ 0 & \text{otherwise} \end{cases}$	$\partial_k^i(y_0, \dots, y_k) = \begin{cases} (y_0, \dots, y_{i-1}, y_{i+1}, \dots, y_k) & \text{if } d(y_{i-1}, y_i) + d(y_i, y_{i+1}) = d(y_{i-1}, y_{i+1}) \\ 0 & \text{otherwise} \end{cases}$
2501.06662_E00001	003	■ 0.822	$\operatorname{Mag}(t \mathcal{M}) = (t-1) \sum_{x \in \operatorname{ob}(\mathcal{M}) \setminus T(\perp)} H_t(p_x) + \#(T(\perp)), \quad t > 0.$	$\operatorname{Mag}(t \mathcal{M}) = (t-1) \sum_{x \in \operatorname{ob}(\mathcal{M}) \setminus T(\perp)} H_t(p_x) + \#(T(\perp)), \quad t > 0.$	$\operatorname{Mag}(t \mathcal{M}) = (t-1) \sum_{x \in \operatorname{ob}(\mathcal{M}) \setminus T(\perp)} H_t(p_x) + \#(T(\perp)), \quad t > 0.$
2501.06662_E00029	016	■ 0.829	$\left(\zeta_t - \delta \right)^k(x, y) = \sum_{\substack{(y_0, y_1, \dots, y_{k-1}, y_k) \\ y_0 = x, y_k = y \\ (y_{i-1}, y_i) \in E \text{ for } i=1, \dots, k}} \prod_{i=1}^k \pi(y_i \mid y_{i-1})^t.$	$(\zeta_t - \delta)^k(x, y) = \sum_{\substack{(y_0, y_1, \dots, y_{k-1}, y_k) \\ y_0 = x, y_k = y \\ (y_{i-1}, y_i) \in E \text{ for } i=1, \dots, k}} \prod_{i=1}^k \pi(y_i \mid y_{i-1})^t$	$(\zeta_t - \delta)^k(x, y) = \sum_{\substack{(y_0, y_1, \dots, y_{k-1}, y_k) \\ y_0 = x, y_k = y \\ (y_{i-1}, y_i) \in E \text{ for } i=1, \dots, k}} \prod_{i=1}^k \pi(y_i \mid y_{i-1})^t.$
2501.06662_E00036 (11)	018	■ 0.835	$\operatorname{Mag}(t \mathcal{M}) = (t-1) \sum_{x \in \operatorname{ob}(\mathcal{M}) \setminus T(\perp)} H_t(p_x) + \#(T(\perp)), \quad t > 0.$	$\operatorname{Mag}(t \mathcal{M}) = (t-1) \sum_{x \in \operatorname{ob}(\mathcal{M}) \setminus T(\perp)} H_t(p_x) + \#(T(\perp)), \quad t > 0.$	$\operatorname{Mag}(t \mathcal{M}) = (t-1) \sum_{x \in \operatorname{ob}(\mathcal{M}) \setminus T(\perp)} H_t(p_x) + \#(T(\perp)), \quad t > 0.$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value					

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2501.06662_E00026 (8)	015	■ 0.842	$\zeta_t^{-1}(x, y) = \sum_{k \geq 0} \sum_{\substack{\text{nondeg. paths} \\ x=y_0 \rightarrow y_1 \rightarrow \dots \rightarrow y_k=y}} (-1)^k \prod_{i=1}^k \pi(y_i y_{i-1})^t.$	$\zeta_t^{-1}(x, y) = \sum_{k \geq 0} \sum_{\substack{\text{nondeg. paths} \\ x=y_0 \rightarrow y_1 \rightarrow \dots \rightarrow y_k=y}} (-1)^k \prod_{i=1}^k \pi(y_i y_{i-1})^t.$	$\zeta_t^{-1}(x, y) = \sum_{k \geq 0} \sum_{\substack{\text{nondeg. paths} \\ x=y_0 \rightarrow y_1 \rightarrow \dots \rightarrow y_k=y}} (-1)^k \prod_{i=1}^k \pi(y_i y_{i-1})^t.$
2501.06662_E00021	014	■ 0.849	$\begin{aligned} \operatorname{Mag}(\mathrm{L}) &= \sum_{x, y \in \mathrm{ob}(\mathrm{L})} \zeta_{\mathrm{L}}^{-1}(x, y) \\ &= \sum_{x, y \in \mathrm{ob}(\mathrm{L})} \mu_{\mathrm{L}}(x, y) \\ &= \sum_{x \in \mathrm{ob}(\mathrm{L})} \mu_{\mathrm{L}}(x, x) + \sum_{\substack{x \perp a \in \mathrm{ob}(\mathrm{L}) \\ a \in \bigcup_{i=0}^{N-2} A^i}} \sum_{a_1 \in A \cup \{\dagger\}} \mu_{\mathrm{L}}(x, xa_1) \\ &= \# \mathrm{ob}(\mathrm{L}) - \# \left\{ x \in \mathrm{ob}(\mathrm{L}) : x = \perp a \text{ with } a \in \bigcup_{i=0}^{N-2} A^i \right\} (\# A + 1). \end{aligned}$	$\begin{aligned} \operatorname{Mag}(\mathrm{L}) &= \sum_{x, y \in \mathrm{ob}(\mathrm{L})} \zeta_{\mathrm{L}}^{-1}(x, y) \\ &= \sum_{x, y \in \mathrm{ob}(\mathrm{L})} \mu_{\mathrm{L}}(x, y) \\ &= \sum_{x \in \mathrm{ob}(\mathrm{L})} \mu_{\mathrm{L}}(x, x) + \sum_{\substack{x \perp a \in \mathrm{ob}(\mathrm{L}) \\ a \in \bigcup_{i=0}^{N-2} A^i}} \sum_{a_1 \in A \cup \{\dagger\}} \mu_{\mathrm{L}}(x, xa_1) \\ &= \# \mathrm{ob}(\mathrm{L}) - \# \left\{ x \in \mathrm{ob}(\mathrm{L}) : x = \perp a \text{ with } a \in \bigcup_{i=0}^{N-2} A^i \right\} (\# A + 1) \end{aligned}$	$\begin{aligned} \operatorname{Mag}(\mathrm{L}) &= \sum_{x, y \in \mathrm{ob}(\mathrm{L})} \zeta_{\mathrm{L}}^{-1}(x, y) \\ &= \sum_{x, y \in \mathrm{ob}(\mathrm{L})} \mu_{\mathrm{L}}(x, y) \\ &= \sum_{x \in \mathrm{ob}(\mathrm{L})} \mu_{\mathrm{L}}(x, x) + \sum_{\substack{x \perp a \in \mathrm{ob}(\mathrm{L}) \\ a \in \bigcup_{i=0}^{N-2} A^i}} \sum_{a_1 \in A \cup \{\dagger\}} \mu_{\mathrm{L}}(x, xa_1) \\ &= \# \mathrm{ob}(\mathrm{L}) - \# \left\{ x \in \mathrm{ob}(\mathrm{L}) : x = \perp a \text{ with } a \in \bigcup_{i=0}^{N-2} A^i \right\} (\# A + 1). \end{aligned}$
2501.06662_E00022	015	■ 0.871	$\begin{aligned} \# \mathrm{ob}(\mathrm{L}) &= 1 + \sum_{i=1}^{N-1} \# \mathrm{L}^{(i)} \\ &= 1 + \sum_{i=1}^{N-1} \# A^i + \# A^{i-1} = \# A^{N-1} + 2(\# A^{N-2}) + \dots + 2(\# A) + 2. \end{aligned}$	$\begin{aligned} \# \mathrm{ob}(\mathrm{L}) &= 1 + \sum_{i=1}^{N-1} \# \mathrm{L}^{(i)} \\ &= 1 + \sum_{i=1}^{N-1} \# A^i + \# A^{i-1} = \# A^{N-1} + 2(\# A^{N-2}) + \dots + 2(\# A) + 2 \end{aligned}$	$\begin{aligned} \# \mathrm{ob}(\mathrm{L}) &= 1 + \sum_{i=1}^{N-1} \# \mathrm{L}^{(i)} \\ &= 1 + \sum_{i=1}^{N-1} \# A^i + \# A^{i-1} = \# A^{N-1} + 2(\# A^{N-2}) + \dots + 2(\# A) + 2. \end{aligned}$
2501.06662_E00045	019	■ 0.922	$\zeta_{\mathcal{M}}^{-1} = \zeta_{\mathcal{M}}^{-1} \big _{\mathrm{ob}(\mathcal{M}_x) \times \mathrm{ob}(\mathcal{M}_x)},$	$\zeta_{\mathcal{M}_x}^{-1} = \zeta_{\mathcal{M}}^{-1} \big _{\mathrm{ob}(\mathcal{M}_x) \times \mathrm{ob}(\mathcal{M}_x)}$	$\zeta_{\mathcal{M}_x}^{-1} = \zeta_{\mathcal{M}}^{-1} \big _{\mathrm{ob}(\mathcal{M}_x) \times \mathrm{ob}(\mathcal{M}_x)},$
2501.06662_E00039 (12)	018	■ 0.940	$f'(1) = \sum_{x \in \mathrm{ob}(\mathcal{M}) \setminus T(\perp)} H(p_x).$	$f'(1) = \sum_{x \in \mathrm{ob}(\mathcal{M}) \setminus T(\perp)} H(p_x).$	$f'(1) = \sum_{x \in \mathrm{ob}(\mathcal{M}) \setminus T(\perp)} H(p_x).$
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2501.06662_E00046	020	■ 0.942	$\begin{aligned} \operatorname{Mag}(\mathcal{M}_x) &= \sum_{y, z \in \operatorname{ob}(\mathcal{M}_x)} \zeta_{\mathcal{M}_x}^{-1}(y, z) \\ &= \sum_{y \in \operatorname{ob}(\mathcal{M}_x)} \zeta_{\mathcal{M}_x}^{-1}(y, y) - \sum_{y \in \operatorname{ob}(\mathcal{M}_x) \setminus T(x)} \sum_{y \in L_y^{(1)}} \pi(z y)^t \\ &= (t-1) \sum_{y \in \operatorname{ob}(\mathcal{M}_x) \setminus T(x)} H_t(p_y) + \#(T(x)). \end{aligned}$	$\operatorname{Mag}(\mathcal{M}_x) = \sum_{y, z \in \operatorname{ob}(\mathcal{M}_x)} \zeta_{\mathcal{M}_x}^{-1}(y, z) \\ = \sum_{y \in \operatorname{ob}(\mathcal{M}_x)} \zeta_{\mathcal{M}_x}^{-1}(y, y) - \sum_{y \in \operatorname{ob}(\mathcal{M}_x) \setminus T(x)} \sum_{y \in L_y^{(1)}} \pi(z y)^t \\ = (t-1) \sum_{y \in \operatorname{ob}(\mathcal{M}_x) \setminus T(x)} H_t(p_y) + \#(T(x)).$	$\operatorname{Mag}(t\mathcal{M}_x) = \sum_{y, z \in \operatorname{ob}(\mathcal{M}_x)} \zeta_{\mathcal{M}_x}^{-1}(y, z) \\ = \sum_{y \in \operatorname{ob}(\mathcal{M}_x)} \zeta_{\mathcal{M}_x}^{-1}(y, y) - \sum_{y \in \operatorname{ob}(\mathcal{M}_x) \setminus T(x)} \sum_{y \in L_y^{(1)}} \pi(z y)^t \\ = (t-1) \sum_{y \in \operatorname{ob}(\mathcal{M}_x) \setminus T(x)} H_t(p_y) + \#(T(x)).$
2501.06662_E00040	019	■ 0.945	$f(t) = \#(\operatorname{ob}(\mathcal{M})) - \sum_{x \in \operatorname{ob}(\mathcal{M})} Z_x(t),$	$f(t) = \#(\operatorname{ob}(\mathcal{M})) - \sum_{x \in \operatorname{ob}(\mathcal{M})} Z_x(t),$	$f(t) = \#(\operatorname{ob}(\mathcal{M})) - \sum_{x \in \operatorname{ob}(\mathcal{M})} Z_x(t),$
2501.06662_E00044	019	■ 0.971	$\zeta_{\mathcal{M}_x} = \zeta_{\mathcal{M}} _{\operatorname{ob}(\mathcal{M}_x) \times \operatorname{ob}(\mathcal{M}_x)}$	$\zeta_{\mathcal{M}_x} = \zeta_{\mathcal{M}} _{\operatorname{ob}(\mathcal{M}_x) \times \operatorname{ob}(\mathcal{M}_x)}$	$\zeta_{\mathcal{M}_x} = \zeta_{\mathcal{M}} _{\operatorname{ob}(\mathcal{M}_x) \times \operatorname{ob}(\mathcal{M}_x)}$
2501.06662_E00032	017	■ 0.973	$\pi(y x)^t \sum_{k \geq 0} (-1)^k \# \{ \text{nondegenerate paths of length } k \text{ from } x \text{ to } y \text{ in } \mathcal{L} \}$	$\pi(y x)^t \sum_{k \geq 0} (-1)^k \# \{ \text{nondegenerate paths of length } k \text{ from } x \text{ to } y \text{ in } \mathcal{L} \}$	$\pi(y x)^t \sum_{k \geq 0} (-1)^k \# \{ \text{nondegenerate paths of length } k \text{ from } x \text{ to } y \text{ in } \mathcal{L} \}$
2501.06662_E00056	022	■ 0.986	$PPL(y) = 1/\zeta_t(a_0, y) = 1/e^{t \ln \pi(y a_0)},$	$PPL(y) = 1/\zeta_t(a_0, y) = 1/e^{t \ln \pi(y a_0)},$	$PPL(y) = 1/\zeta_t(a_0, y) = 1/e^{t \ln \pi(y a_0)},$
2501.06662_E00015 (7)	012	■ 0.988	$\operatorname{Mag}(\mathcal{C}) = \sum_{(x, y) \in \operatorname{ob}(\mathcal{C}) \times \operatorname{ob}(\mathcal{C})} \zeta_{\mathcal{C}}^{-1}(x, y).$	$\operatorname{Mag}(\mathcal{C}) = \sum_{(x, y) \in \operatorname{ob}(\mathcal{C}) \times \operatorname{ob}(\mathcal{C})} \zeta_{\mathcal{C}}^{-1}(x, y).$	$\operatorname{Mag}(\mathcal{C}) = \sum_{(x, y) \in \operatorname{ob}(\mathcal{C}) \times \operatorname{ob}(\mathcal{C})} \zeta_{\mathcal{C}}^{-1}(x, y).$
2501.06662_E00002	004	■ 0.991	$\frac{(t-1)(n^{1-t}-1)}{1-t} \cdot \#(\operatorname{ob}(\mathcal{M}) \setminus T(\perp)) + \#(T(\perp))$	$\frac{(t-1)(n^{1-t}-1)}{1-t} \cdot \#(\operatorname{ob}(\mathcal{M}) \setminus T(\perp)) + \#(T(\perp))$	$\frac{(t-1)(n^{1-t}-1)}{1-t} \cdot \#(\operatorname{ob}(\mathcal{M}) \setminus T(\perp)) + \#(T(\perp))$
2501.06662_E00005	006	■ 0.991	$\operatorname{ob}(\mathcal{L}) = \{ \perp a : a \in A^* \text{ and } a \leq N-1 \} \sqcup \{ \perp a^\dagger : a \in A^* \text{ and } a < N-1 \}.$	$\operatorname{ob}(\mathcal{L}) = \{ \perp a : a \in A^* \text{ and } a \leq N-1 \} \sqcup \{ \perp a^\dagger : a \in A^* \text{ and } a < N-1 \}.$	$\operatorname{ob}(\mathcal{L}) = \{ \perp a : a \in A^* \text{ and } a \leq N-1 \} \sqcup \{ \perp a^\dagger : a \in A^* \text{ and } a < N-1 \}.$
2501.06662_E00006	008	■ 0.996	$T(x) = \{ y \in \operatorname{ob}(\mathcal{L}_x) : y \text{ is an unfinished text of length } N \text{ or } y \text{ is a finished text such that } y \leq N \}.$	$T(x) = \{ y \in \operatorname{ob}(\mathcal{L}_x) : y \text{ is an unfinished text of length } N \text{ or } y \text{ is a finished text such that } y \leq N \}.$	$T(x) = \{ y \in \operatorname{ob}(\mathcal{L}_x) : y \text{ is an unfinished text of length } N \text{ or } y \text{ is a finished text such that } y \leq N \}.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value					

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2501.06662_E00034	017	■ 0.998	$\operatorname{Mag}(t \mathcal{M}) := \sum_{x, y \in \operatorname{ob}(\mathcal{M})} \zeta_t^{-1}(x, y)$	$\operatorname{Mag}(t \mathcal{M}) := \sum_{x, y \in \operatorname{ob}(\mathcal{M})} \zeta_t^{-1}(x, y).$	$\operatorname{Mag}(t \mathcal{M}) := \sum_{x, y \in \operatorname{ob}(\mathcal{M})} \zeta_t^{-1}(x, y).$
2501.06662_E00003	004	■ 0.998	$\underbrace{\#(T(\perp))}_{\text{deterministic LM}} \leq \lim_{t \rightarrow \infty} \operatorname{Mag}(t \mathcal{M}) \leq \underbrace{\#(\operatorname{ob}(\mathcal{M}))}_{\text{maximally random LM}}.$	$\underbrace{\#(T(\perp))}_{\text{deterministic LM}} \leq \lim_{t \rightarrow \infty} \operatorname{Mag}(t \mathcal{M}) \leq \underbrace{\#(\operatorname{ob}(\mathcal{M}))}_{\text{maximally random LM}}.$	$\underbrace{\#(T(\perp))}_{\text{deterministic LM}} \leq \lim_{t \rightarrow \infty} \operatorname{Mag}(t \mathcal{M}) \leq \underbrace{\#(\operatorname{ob}(\mathcal{M}))}_{\text{maximally random LM}}.$
2501.06662_E00054 (14)	022	■ 0.998	$\operatorname{Mag}(t \mathcal{M}) = \operatorname{rank} H_{0,0}(\mathcal{M}) - \sum_{\ell \geq 0} q^\ell \operatorname{rank} H_{1,\ell}(\mathcal{M}).$	$\operatorname{Mag}(t \mathcal{M}) = \operatorname{rank} H_{0,0}(\mathcal{M}) - \sum_{\ell \geq 0} q^\ell \operatorname{rank} H_{1,\ell}(\mathcal{M}).$	$\operatorname{Mag}(t \mathcal{M}) = \operatorname{rank} H_{0,0}(\mathcal{M}) - \sum_{\ell \geq 0} q^\ell \operatorname{rank} H_{1,\ell}(\mathcal{M}).$
2501.06662_E00019	014	■ 0.999	$\mu_P(s, u) = \begin{cases} 1 & \text{if } s = u \\ -\sum_{s \leq t < u} \mu_P(s, t) & \text{if } s < u \\ 0 & \text{otherwise} \end{cases}$	$\mu_P(s, u) = \begin{cases} 1 & \text{if } s = u \\ -\sum_{s \leq t < u} \mu_P(s, t) & \text{if } s < u \\ 0 & \text{otherwise} \end{cases}.$	$\mu_P(s, u) = \begin{cases} 1 & \text{if } s = u \\ -\sum_{s \leq t < u} \mu_P(s, t) & \text{if } s < u \\ 0 & \text{otherwise.} \end{cases}$
2501.06662_E00042	019	■ 0.999	$H(p_x) = -\left. \frac{d}{dt} Z_x(t) \right _{t=1},$	$H(p_x) = -\left. \frac{d}{dt} Z_x(t) \right _{t=1},$	$H(p_x) = -\left. \frac{d}{dt} Z_x(t) \right _{t=1},$
2501.06662_E00004	005	■ 1.000	$\operatorname{Mag}(t \mathcal{M}) = \sum_{\ell} e^{-t\ell} \sum_{k \geq 0} (-1)^k \operatorname{rank}(H_{k,\ell}(\mathcal{M})),$	$\operatorname{Mag}(t \mathcal{M}) = \sum_{\ell} e^{-t\ell} \sum_{k \geq 0} (-1)^k \operatorname{rank}(H_{k,\ell}(\mathcal{M})),$	$\operatorname{Mag}(t \mathcal{M}) = \sum_{\ell} e^{-t\ell} \sum_{k \geq 0} (-1)^k \operatorname{rank}(H_{k,\ell}(\mathcal{M})),$
2501.06662_E00012	011	■ 1.000	$\begin{aligned} & x = \perp a_1 \cdots a_t \\ & y = \perp a_1 \cdots a_t \cdots a_{t+k} \\ & z = \perp a_1 \cdots a_t \cdots a_{t+k} \cdots a_{t+k+k'} \end{aligned}$	$\begin{aligned} x &= \perp a_1 \cdots a_t \\ y &= \perp a_1 \cdots a_t \cdots a_{t+k} \\ z &= \perp a_1 \cdots a_t \cdots a_{t+k} \cdots a_{t+k+k'} \end{aligned}$	$\begin{aligned} x &= \perp a_1 \cdots a_t \\ y &= \perp a_1 \cdots a_t \cdots a_{t+k} \\ z &= \perp a_1 \cdots a_t \cdots a_{t+k} \cdots a_{t+k+k'}. \end{aligned}$
2501.06662_E00031	017	■ 1.000	$\begin{aligned} & \prod_{i=1}^k \pi(y_i y_{i-1})^t = \pi(y_1 y_0)^t \pi(y_2 y_1)^t \pi(y_3 y_2)^t \cdots \pi(y_k y_{k-1})^t \\ & = \pi(y_2 y_0)^t \pi(y_3 y_2)^t \cdots \pi(y_k y_{k-1})^t \\ & = \vdots \\ & = \pi(y_k y_0)^t \end{aligned}$	$\begin{aligned} \prod_{i=1}^k \pi(y_i y_{i-1})^t &= \pi(y_1 y_0)^t \pi(y_2 y_1)^t \pi(y_3 y_2)^t \cdots \pi(y_k y_{k-1})^t \\ &= \pi(y_2 y_0)^t \pi(y_3 y_2)^t \cdots \pi(y_k y_{k-1})^t \\ &= \vdots \\ &= \pi(y_k y_0)^t \end{aligned}$	$\begin{aligned} \prod_{i=1}^k \pi(y_i y_{i-1})^t &= \pi(y_1 y_0)^t \pi(y_2 y_1)^t \pi(y_3 y_2)^t \cdots \pi(y_k y_{k-1})^t \\ &= \pi(y_2 y_0)^t \pi(y_3 y_2)^t \cdots \pi(y_k y_{k-1})^t \\ &= \vdots \\ &= \pi(y_k y_0)^t. \end{aligned}$

Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
2501.06662_E00007	008	■ 1.000	$\mathrm{p}\left(\mathrm{a}_{\mathrm{3}}\mid\mathrm{perp}\mathrm{a}\right)\mathrm{p}\left(\mathrm{a}_{\mathrm{4}}\mid\mathrm{perp}\mathrm{a}\mathrm{a}_{\mathrm{3}}\right)\mathrm{p}\left(\mathrm{a}_{\mathrm{5}}\mid\mathrm{perp}\mathrm{a}\mathrm{a}_{\mathrm{3}}\mathrm{a}_{\mathrm{4}}\right)\mathrm{p}\left(\dagger\mid\mathrm{perp}\mathrm{a}\mathrm{a}_{\mathrm{3}}\mathrm{a}_{\mathrm{4}}\mathrm{a}_{\mathrm{5}}\right).$	$p(a_3\mid a)p(a_4\mid aa_3)p(a_5\mid aa_3a_4)p(\dagger\mid aa_3a_4a_5).$	$p(a_3\mid a)p(a_4\mid aa_3)p(a_5\mid aa_3a_4)p(\dagger\mid aa_3a_4a_5).$
2501.06662_E00008 (1)	009	■ 1.000	$\mathrm{\pi}(y\mid x):=\left\{\begin{array}{l} 1 \\ 0 \\ \prod_{i=1}^kp(a_{t+i}\mid y_{<t+i}) \end{array}\right.\begin{array}{l} \text{if } x=y \\ \text{if } x\nrightarrow y \\ \text{if } x\rightarrow y \end{array}.$	$\pi(y\mid x):=\left\{\begin{array}{l} 1 \\ 0 \\ \prod_{i=1}^kp(a_{t+i}\mid y_{<t+i}) \end{array}\right.\begin{array}{l} \text{if } x=y \\ \text{if } x\nrightarrow y \\ \text{if } x\rightarrow y \end{array}.$	$\pi(y x):=\left\{\begin{array}{l} 1 \\ 0 \\ \prod_{i=1}^kp(a_{t+i} y_{<t+i}) \end{array}\right.\begin{array}{l} \text{if } x=y \\ \text{if } x\nrightarrow y \\ \text{if } x\rightarrow y \end{array}.$
2501.06662_E00009 (2)	009	■ 1.000	$\sum_{y\in T(x)}\mathrm{\pi}(y\mid x)=\sum_{a\in A\cup\{\dagger\}}p_x(a)=1,$	$\sum_{y\in T(x)}\pi(y\mid x)=\sum_{a\in A\cup\{\dagger\}}p_x(a)=1,$	$\sum_{y\in T(x)}\pi(y x)=\sum_{a\in A\cup\{\dagger\}}p_x(a)=1,$
2501.06662_E00010 (4)	009	■ 1.000	$\begin{aligned} \sum_{i=0}^{m-1}\sum_{a'\in A^i}\mathrm{\pi}(xa'\dagger\mid x)+\sum_{a\in A^m}\mathrm{\pi}(xa\mid x) \\ =\sum_{i=0}^{m-1}\sum_{a'\in A^i}\mathrm{\pi}(xa'\dagger\mid x)+\sum_{\substack{a=a'a'' \\ a'\in A^{m-1},a''\in A}}p(a''\mid xa')\mathrm{\pi}(xa'\mid x) \\ =\sum_{i=0}^{m-2}\sum_{a'\in A^i}\mathrm{\pi}(xa'\dagger\mid x)+\sum_{a'\in A^{m-1}}\mathrm{\pi}(xa'\mid x)\sum_{a''\in A\cup\{\dagger\}}p(a''\mid xa') \\ =\sum_{i=0}^{m-2}\sum_{a'\in A^i}\mathrm{\pi}(xa'\dagger\mid x)+\sum_{a'\in A^{m-1}}\mathrm{\pi}(xa'\mid x) \end{aligned}$	$\begin{aligned} \sum_{y\in T(x)}\pi(y\mid x) &= \sum_{i=0}^{m-1}\sum_{a'\in A^i}\pi(xa'\dagger\mid x) + \sum_{a\in A^m}\pi(xa\mid x) \\ &= \sum_{i=0}^{m-1}\sum_{a'\in A^i}\pi(xa'\dagger\mid x) + \sum_{\substack{a=a'a'' \\ a'\in A^{m-1}, a''\in A}}p(a''\mid xa')\pi(xa'\mid x) \\ &= \sum_{i=0}^{m-2}\sum_{a'\in A^i}\pi(xa'\dagger\mid x) + \sum_{a'\in A^{m-1}}\pi(xa'\mid x)\sum_{a''\in A\cup\{\dagger\}}p(a''\mid xa') \\ &= \sum_{i=0}^{m-2}\sum_{a'\in A^i}\pi(xa'\dagger\mid x) + \sum_{a'\in A^{m-1}}\pi(xa'\mid x) \end{aligned}$	$\begin{aligned} \sum_{y\in T(x)}\pi(y x) &= \sum_{i=0}^{m-1}\sum_{a'\in A^i}\pi(xa'\dagger\mid x) + \sum_{a\in A^m}\pi(xa\mid x) \\ &= \sum_{i=0}^{m-1}\sum_{a'\in A^i}\pi(xa'\dagger\mid x) + \sum_{\substack{a=a'a'' \\ a'\in A^{m-1}, a''\in A}}p(a''\mid xa')\pi(xa'\mid x) \\ &= \sum_{i=0}^{m-2}\sum_{a'\in A^i}\pi(xa'\dagger\mid x) + \sum_{a'\in A^{m-1}}\pi(xa'\mid x)\sum_{a''\in A\cup\{\dagger\}}p(a''\mid xa') \\ &= \sum_{i=0}^{m-2}\sum_{a'\in A^i}\pi(xa'\dagger\mid x) + \sum_{a'\in A^{m-1}}\pi(xa'\mid x) \end{aligned}$
2501.06662_E00011	010	■ 1.000	$\begin{aligned} 1 &\leq \mathcal{C}(x, x) \\ \mathcal{C}(y, z) &\otimes \mathcal{C}(x, y) \leq \mathcal{C}(x, z) \end{aligned}$	$\begin{aligned} 1 &\leq \mathcal{C}(x, x) \\ \mathcal{C}(y, z) \otimes \mathcal{C}(x, y) &\leq \mathcal{C}(x, z) \end{aligned}$	$\begin{aligned} 1 &\leq \mathcal{C}(x, x) \\ \mathcal{C}(y, z) \otimes \mathcal{C}(x, y) &\leq \mathcal{C}(x, z) \end{aligned}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
2501.06662_E00013 (6)	011	■ 1.000	$\begin{aligned} \pi(y \mid x) \pi(z \mid y) &= \prod_{i=1}^k p(a_{t+i} \mid y_{<t+i}) \prod_{j=1}^{k'} p(a_{t+k+j} \mid z_{<t+k+j}) \\ &= \prod_{i=1}^{k+k'} p(a_{t+i} \mid z_{<t+i}) \\ &= \pi(z \mid x). \end{aligned}$	$\pi(y \mid x) \pi(z \mid y) = \prod_{i=1}^k p(a_{t+i} \mid y_{<t+i}) \prod_{j=1}^{k'} p(a_{t+k+j} \mid z_{<t+k+j})$ $= \prod_{i=1}^{k+k'} p(a_{t+i} \mid z_{<t+i})$ $= \pi(z \mid x)$	$\pi(y x)\pi(z y) = \prod_{i=1}^k p(a_{t+i} y_{<t+i}) \prod_{j=1}^{k'} p(a_{t+k+j} z_{<t+k+j})$ $= \prod_{i=1}^{k+k'} p(a_{t+i} z_{<t+i})$ $= \pi(z x).$
2501.06662_E00014	012	■ 1.000	$d(x, y) := -\ln \pi(y \mid x) \text{ .}$	$d(x,y) := -\ln \pi(y \mid x).$	$d(x,y) := -\ln \pi(y x).$
2501.06662_E00016	013	■ 1.000	$(f * g)(s, u) := \sum_{s \leq t \leq u} f(s, t) g(t, u),$	$(f * g)(s, u) := \sum_{s \leq t \leq u} f(s, t) g(t, u),$	$(f * g)(s, u) := \sum_{s \leq t \leq u} f(s, t) g(t, u),$
2501.06662_E00017	013	■ 1.000	$\delta(s, u) = \begin{cases} 1 & \text{if } s = u \\ 0 & \text{if } s \neq u \end{cases}$	$\delta(s, u) = \begin{cases} 1 & \text{if } s = u \\ 0 & \text{if } s \neq u \end{cases}$	$\delta(s, u) = \begin{cases} 1 & \text{if } s = u \\ 0 & \text{if } s \neq u \end{cases}$
2501.06662_E00018	013	■ 1.000	$\zeta_P(s, u) = 1, \quad \text{for all } s \leq u \text{ in } P.$	$\zeta_P(s, u) = 1, \quad \text{for all } s \leq u \text{ in } P.$	$\zeta_P(s, u) = 1, \quad \text{for all } s \leq u \text{ in } P.$
2501.06662_E00020	014	■ 1.000	$\begin{aligned} \mu_{\mathrm{L}}(x, x) &= 1 \\ \mu_{\mathrm{L}}(x, xa_1) &= -\mu_{\mathrm{L}}(x, x) = -1 \\ \mu_{\mathrm{L}}(x, xa_1a_2) &= -\mu_{\mathrm{L}}(x, x) - \mu_{\mathrm{L}}(x, xa_1) = -1 + 1 = 0 \\ \mu_{\mathrm{L}}(x, xa_1a_2a_3) &= -\mu_{\mathrm{L}}(x, x) - \mu_{\mathrm{L}}(x, xa_1) - \mu_{\mathrm{L}}(x, xa_1a_2) = -1 + 1 + 0 = 0. \end{aligned}$	$\begin{aligned} \mu_{\mathrm{L}}(x, x) &= 1 \\ \mu_{\mathrm{L}}(x, xa_1) &= -\mu_{\mathrm{L}}(x, x) = -1 \\ \mu_{\mathrm{L}}(x, xa_1a_2) &= -\mu_{\mathrm{L}}(x, x) - \mu_{\mathrm{L}}(x, xa_1) = -1 + 1 = 0 \\ \mu_{\mathrm{L}}(x, xa_1a_2a_3) &= -\mu_{\mathrm{L}}(x, x) - \mu_{\mathrm{L}}(x, xa_1) - \mu_{\mathrm{L}}(x, xa_1a_2) = -1 + 1 + 0 = 0. \end{aligned}$	$\begin{aligned} \mu_{\mathrm{L}}(x, x) &= 1 \\ \mu_{\mathrm{L}}(x, xa_1) &= -\mu_{\mathrm{L}}(x, x) = -1 \\ \mu_{\mathrm{L}}(x, xa_1a_2) &= -\mu_{\mathrm{L}}(x, x) - \mu_{\mathrm{L}}(x, xa_1) = -1 + 1 = 0 \\ \mu_{\mathrm{L}}(x, xa_1a_2a_3) &= -\mu_{\mathrm{L}}(x, x) - \mu_{\mathrm{L}}(x, xa_1) - \mu_{\mathrm{L}}(x, xa_1a_2) = -1 + 1 + 0 = 0. \end{aligned}$
2501.06662_E00024	015	■ 1.000	$\zeta_{\mathcal{M}}(x, y) := e^{-d(x, y)} = \pi(y \mid x),$	$\zeta_{\mathcal{M}}(x, y) := e^{-d(x, y)} = \pi(y \mid x),$	$\zeta_{\mathcal{M}}(x, y) := e^{-d(x, y)} = \pi(y x),$
2501.06662_E00025	015	■ 1.000	$\left(\zeta_{\mathcal{M}} \right)_t(x, y) := e^{-td(x, y)} = \pi(y \mid x)^t.$	$(\zeta_{\mathcal{M}})_t(x, y) := e^{-td(x, y)} = \pi(y \mid x)^t.$	$(\zeta_{\mathcal{M}})_t(x, y) := e^{-td(x, y)} = \pi(y x)^t.$
2501.06662_E00027 (9)	016	■ 1.000	$\zeta_t^{-1} = \sum_{k \geq 0} (-1)^k (\zeta_t - \delta)^k$	$\zeta_t^{-1} = \sum_{k \geq 0} (-1)^k (\zeta_t - \delta)^k$	$\zeta_t^{-1} = \sum_{k \geq 0} (-1)^k (\zeta_t - \delta)^k,$
2501.06662_E00028	016	■ 1.000	$(\zeta_t - \delta)(x, y) = \pi(x, y)^t \mathbf{1}_{(x, y) \in E},$	$(\zeta_t - \delta)(x, y) = \pi(x, y)^t \mathbf{1}_{(x, y) \in E},$	$(\zeta_t - \delta)(x, y) = \pi(x, y)^t \mathbf{1}_{(x, y) \in E},$
2501.06662_E00030	016	■ 1.000	$\zeta_t^{-1}(x, y) = \pi(y \mid x)^t \zeta_{\mathrm{L}}^{-1}(x, y).$	$\zeta_t^{-1}(x, y) = \pi(y \mid x)^t \zeta_{\mathrm{L}}^{-1}(x, y).$	$\zeta_t^{-1}(x, y) = \pi(y x)^t \zeta_{\mathrm{L}}^{-1}(x, y).$
2501.06662_E00035	017	■ 1.000	$H_t(p) = \frac{1}{t-1} \left(1 - \sum_{i=1}^n p_i^t \right).$	$H_t(p) = \frac{1}{t-1} \left(1 - \sum_{i=1}^n p_i^t \right).$	$H_t(p) = \frac{1}{t-1} \left(1 - \sum_{i=1}^n p_i^t \right).$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
2501.06662_E00038	018	■ 1.000	$\lim_{t \rightarrow 1} H_t(p) = \lim_{t \rightarrow 1} \frac{1}{t-1} \left(1 - \sum_{s \in S} p(s)^t \right) = - \sum_{s \in S} p(s) \ln p(s) =: H(p).$	$\lim_{t \rightarrow 1} H_t(p) = \lim_{t \rightarrow 1} \frac{1}{t-1} \left(1 - \sum_{s \in S} p(s)^t \right) = - \sum_{s \in S} p(s) \ln p(s) =: H(p).$	$\lim_{t \rightarrow 1} H_t(p) = \lim_{t \rightarrow 1} \frac{1}{t-1} \left(1 - \sum_{s \in S} p(s)^t \right) = - \sum_{s \in S} p(s) \ln p(s) =: H(p).$
2501.06662_E00041	019	■ 1.000	$Z_x(t) = \sum_{a \in A \cup \{\dagger\}} p_x(a)^t = \sum_{a \in A \cup \{\dagger\}} e^{-t(-\ln p_x(a))}$	$Z_x(t) = \sum_{a \in A \cup \{\dagger\}} p_x(a)^t = \sum_{a \in A \cup \{\dagger\}} e^{-t(-\ln p_x(a))}$	$Z_x(t) = \sum_{a \in A \cup \{\dagger\}} p_x(a)^t = \sum_{a \in A \cup \{\dagger\}} e^{-t(-\ln p_x(a))}$
2501.06662_E00043	019	■ 1.000	$\mathbb{E}_\rho(E) = \sum_{s \in S} E(s) \frac{e^{-tE(s)}}{\tilde{Z}(t)} = -\frac{d}{dt} \ln \tilde{Z}(t) = \frac{f'(t)}{\tilde{Z}(t)},$	$\mathbb{E}_\rho(E) = \sum_{s \in S} E(s) \frac{e^{-tE(s)}}{\tilde{Z}(t)} = -\frac{d}{dt} \ln \tilde{Z}(t) = \frac{f'(t)}{\tilde{Z}(t)},$	$\mathbb{E}_\rho(E) = \sum_{s \in S} E(s) \frac{e^{-tE(s)}}{\tilde{Z}(t)} = -\frac{d}{dt} \ln \tilde{Z}(t) = \frac{f'(t)}{\tilde{Z}(t)},$
2501.06662_E00047	020	■ 1.000	$G_{k,\ell} = \left\{ (y_0, \dots, y_k) \in M^{k+1} \mid \sum_{i=0}^{k-1} d(y_i, y_{i+1}) = \ell \text{ and for all } i, y_i \neq y_{i+1} \right\}.$	$G_{k,\ell} = \left\{ (y_0, \dots, y_k) \in M^{k+1} \mid \sum_{i=0}^{k-1} d(y_i, y_{i+1}) = \ell \text{ and for all } i, y_i \neq y_{i+1} \right\}.$	$G_{k,\ell} = \left\{ (y_0, \dots, y_k) \in M^{k+1} \mid \sum_{i=0}^{k-1} d(y_i, y_{i+1}) = \ell \text{ and for all } i, y_i \neq y_{i+1} \right\}.$
2501.06662_E00048	020	■ 1.000	$\partial_k = \sum_{i=0}^k (-1)^i \partial_k^i$	$\partial_k = \sum_{i=0}^k (-1)^i \partial_k^i$	$\partial_k = \sum_{i=0}^k (-1)^i \partial_k^i$
2501.06662_E00050	021	■ 1.000	$\text{Mag}(t\mathcal{M}) = \sum_{\ell} q^\ell \sum_{k \geq 0} (-1)^k \text{rank}(H_{k,\ell}(\mathcal{M}))$	$\text{Mag}(t\mathcal{M}) = \sum_{\ell} q^\ell \sum_{k \geq 0} (-1)^k \text{rank}(H_{k,\ell}(\mathcal{M}))$	$\text{Mag}(t\mathcal{M}) = \sum_{\ell} q^\ell \sum_{k \geq 0} (-1)^k \text{rank}(H_{k,\ell}(\mathcal{M}))$
2501.06662_E00053	021	■ 1.000	$\sum_{c \in G_{k,\ell}} (-1)^k e^{-t\ell} = (-1)^k q^\ell \#G_{k,\ell} = (-1)^k q^\ell \text{rank}(MC_{k,\ell}(\mathcal{M})).$	$\sum_{c \in G_{k,\ell}} (-1)^k e^{-t\ell} = (-1)^k q^\ell \#G_{k,\ell} = (-1)^k q^\ell \text{rank}(MC_{k,\ell}(\mathcal{M})).$	$\sum_{c \in G_{k,\ell}} (-1)^k e^{-t\ell} = (-1)^k q^\ell \#G_{k,\ell} = (-1)^k q^\ell \text{rank}(MC_{k,\ell}(\mathcal{M})).$
2501.06662_E00055	022	■ 1.000	$PPL(x) = \exp \left\{ -\frac{1}{n} \sum_{i=1}^n \ln p(a_i y_{<i}) \right\}$	$PPL(x) = \exp \left\{ -\frac{1}{n} \sum_{i=1}^n \ln p(a_i y_{<i}) \right\}$	$PPL(x) = \exp \left\{ -\frac{1}{n} \sum_{i=1}^n \ln p(a_i y_{<i}) \right\}$
2501.06662_E00057	023	■ 1.000	$\theta(a, b) = 0 \quad \text{if} \quad A(a, b) = \emptyset.$	$\theta(a, b) = 0 \quad \text{if} \quad A(a, b) = \emptyset.$	$\theta(a, b) = 0 \quad \text{if} \quad A(a, b) = \emptyset.$
2501.06662_E00060	023	■ 1.000	$-\sum_{a \in A} \sum_{a' \in A} p_{\perp}(a) P(a, a') \log P(a, a'),$	$-\sum_{a \in A} \sum_{a' \in A} p_{\perp}(a) P(a, a') \log P(a, a'),$	$-\sum_{a \in A} \sum_{a' \in A} p_{\perp}(a) P(a, a') \log P(a, a'),$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value					

Inline formulas (first occurrence)

Identifier	Page	Conf.	LaTeX source	Rendered
2501.06662_F00001	004	—	t	t
2501.06662_F00002	001	—	$\pi(y \mid x) \in [0,1]$	$\pi(y \mid x) \in [0,1]$
2501.06662_F00003	001	—	x	x
2501.06662_F00004	001	—	y	y
2501.06662_F00005	002	—	$\pi(y \mid x)$	$\pi(y \mid x)$
2501.06662_F00006	002	—	n	n
2501.06662_F00007	002	—	(n-1)	$(n-1)$
2501.06662_F00008	002	—	$\dagger$	$\dagger$
2501.06662_F00009	002	—	a	a
2501.06662_F00010	002	—	$x=\perp a$	$x=\perp a$
2501.06662_F00011	002	—	p_x	p_x
2501.06662_F00012	002	—	a_1	a_1
2501.06662_F00013	002	—	$x\,a_1$	xa_1
2501.06662_F00014	002	—	p_{xa}	p_{xa}
2501.06662_F00015	002	—	a_2	a_2
2501.06662_F00016	002	—	$a_2=\dagger$	$a_2=\dagger$
2501.06662_F00017	002	—	aa_1	aa_1
2501.06662_F00018	002	—	π	π
2501.06662_F00019	003	—	$\pi(x \mid x)=1$	$\pi(x \mid x)=1$
2501.06662_F00020	003	—	$\pi(- \mid x)$	$\pi(- \mid x)$
2501.06662_F00021	003	—	$T(x)$	$T(x)$
2501.06662_F00022	003	—	$[0,\infty]$	$[0,\infty]$
2501.06662_F00023	003	—	$-\ln \pi(y \mid x)$	$-\ln \pi(y \mid x)$
2501.06662_F00024	003	—	$([0,\infty],\geq,+,\emptyset)$	$([0,\infty],\geq,+,\emptyset)$
2501.06662_F00025	003	—	$\mathcal{M}$	$\mathcal{M}$
2501.06662_F00026	003	—	$x,y,\ldots$	$x,y,\ldots$
2501.06662_F00027	003	—	$d(x,y)=-\ln \pi(y \mid x)$	$d(x,y)=-\ln \pi(y \mid x)$
2501.06662_F00028	003	—	$\operatorname{Mag}(t\,\mathcal{M})$	$\operatorname{Mag}(t\mathcal{M})$
2501.06662_F00029	003	—	$t>0$	$t>0$
2501.06662_F00030	004	—	$\#(T(\perp))$	$\#(T(\perp))$
2501.06662_F00031	004	—	H_t	H_t
2501.06662_F00032	004	—	$H_t(p_1,\ldots,p_m)=\left(1-\sum_{i=1}^m p_i^t\right)/(t-1)$	$H_t(p_1,\ldots,p_m)=\left(1-\sum_{i=1}^m p_i^t\right)/(t-1)$
2501.06662_F00033	004	—	$(H_t)_{t\in(0,\infty)\setminus\{1\}}$	$(H_t)_{t\in(0,\infty)\setminus\{1\}}$
2501.06662_F00034	004	—	$t>1$	$t>1$
2501.06662_F00035	004	—	$H_t(p_x)$	$H_t(p_x)$
2501.06662_F00036	004	—	$(n^{1-t}-1)/(1-t)$	$(n^{1-t}-1)/(1-t)$
2501.06662_F00037	004	—	$t\rightarrow\infty$	$t\rightarrow\infty$
2501.06662_F00038	004	—	$\mathcal{M}_1$	$\mathcal{M}_1$
2501.06662_F00039	004	—	$\mathcal{M}_2$	$\mathcal{M}_2$
2501.06662_F00040	004	—	$p_{\bullet}^{(1)}$	$p_{\bullet}^{(1)}$
2501.06662_F00041	004	—	$p_{\bullet}^{(2)}$	$p_{\bullet}^{(2)}$
2501.06662_F00042	004	—	x'	x'
2501.06662_F00043	004	—	$H_t(p_{x'}^{(1)})>H_t(p_x^{(2)})$	$H_t\left(p_{x'}^{(1)}\right)>H_t\left(p_x^{(2)}\right)$
2501.06662_F00044	004	—	$\operatorname{Mag}(\left(t\,\mathcal{M}_1\right))>\operatorname{Mag}(\left(t\,\mathcal{M}_2\right))$	$\operatorname{Mag}(t\mathcal{M}_1)>\operatorname{Mag}(t\mathcal{M}_2)$
2501.06662_F00045	004	—	$t<1$	$t<1$
2501.06662_F00046	004	—	$\lim_{t\rightarrow 0}\operatorname{Mag}(t\,\mathcal{M})=(1-n)\cdot\#\left(\operatorname{ob}(\mathcal{M})\backslash T(\perp)\right)+\#(T(\perp))$	$\lim_{t\rightarrow 0}\operatorname{Mag}(t\mathcal{M})=(1-n)\cdot\#\left(\operatorname{ob}(\mathcal{M})\backslash T(\perp)\right)+\#(T(\perp))$
2501.06662_F00047	004	—	$n>1$	$n>1$
2501.06662_F00048	004	—	$t=1$	$t=1$
2501.06662_F00049	004	—	$\sum_{x\in\operatorname{ob}(\mathcal{M})\backslash T(\perp)}H(p_x)$	$\sum_{x\in\operatorname{ob}(\mathcal{M})\backslash T(\perp)}H(p_x)$
2501.06662_F00050	004	—	$\mathcal{M}_x$	$\mathcal{M}_x$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value				

Identifier	Page	Conf.	LaTeX source	Rendered
2501.06662_F00051	004	—	$\left(\sum_y \mu(x, y)\right)_x$	$\left(\sum_y \mu(x, y)\right)_x$
2501.06662_F00052	005	—	μ	μ
2501.06662_F00053	005	—	ℓ	ℓ
2501.06662_F00054	005	—	$\mathcal{L}$	$\mathcal{L}$
2501.06662_F00055	005	—	$-\ln : [0, 1] \rightarrow [0, \infty]$	$-\ln : [0, 1] \rightarrow [0, \infty]$
2501.06662_F00056	006	—	A	A
2501.06662_F00057	006	—	$a, b, \ldots$	$a, b, \dots$
2501.06662_F00058	006	—	$A^{\ast}$	$A^{\ast}$
2501.06662_F00059	006	—	$a \leq b$	$a \leq b$
2501.06662_F00060	006	—	b	b
2501.06662_F00061	006	—	$b{=}a\,a^{\prime}$	$b = aa'$
2501.06662_F00062	006	—	$a^{\prime} \in A^{\ast}$	$a' \in A^{\ast}$
2501.06662_F00063	006	—	$\leq$	$\leq$
2501.06662_F00064	006	—	ϵ	ϵ
2501.06662_F00065	006	—	$ a $	$ a $
2501.06662_F00066	006	—	$a \in A^{\ast}$	$a \in A^{\ast}$
2501.06662_F00067	006	—	$(P, \leq)$	$(P, \leq)$
2501.06662_F00068	006	—	P	P
2501.06662_F00069	006	—	$x \rightarrow y$	$x \rightarrow y$
2501.06662_F00070	006	—	$x \leq y$	$x \leq y$
2501.06662_F00071	006	—	$N \in \mathbb{N}$	$N \in \mathbb{N}$
2501.06662_F00072	006	—	$\mathrm{L} := \mathrm{L}^{\leq N}$	$\mathrm{L} := \mathrm{L}^{\leq N}$
2501.06662_F00073	006	—	$\left((A \cup \{\perp, \dagger\})^{\ast}, \leq\right)$	$((A \cup \{\perp, \dagger\})^{\ast}, \leq)$
2501.06662_F00074	006	—	$N-1$	$N-1$
2501.06662_F00075	006	—	x, y	x, y
2501.06662_F00076	006	—	$(A \cup \{\perp, \dagger\})^{\ast}$	$(A \cup \{\perp, \dagger\})^{\ast}$
2501.06662_F00077	006	—	$\perp a$	$\perp a$
2501.06662_F00078	006	—	$\perp a \dagger$	$\perp a \dagger$
2501.06662_F00079	006	—	$ x $	$ x $
2501.06662_F00080	006	—	$ \perp =1$	$ \perp =1$
2501.06662_F00081	006	—	$\operatorname{ob}(\mathrm{L})$	$\operatorname{ob}(\mathrm{L})$
2501.06662_F00082	006	—	$\bigsqcup_{j=0}^{N-1} \mathrm{L}^{\sim(j)}$	$\bigsqcup_{j=0}^{N-1} \mathrm{L}^{(j)}$
2501.06662_F00083	006	—	$\mathrm{L}^{\sim(j)}$	$\mathrm{L}^{(j)}$
2501.06662_F00084	006	—	$1+j$	$1+j$
2501.06662_F00085	006	—	j	j
2501.06662_F00086	006	—	$L^{\sim(j)}$	$L^{(j)}$
2501.06662_F00087	006	—	$\mathrm{L}^{\sim(i)}$	$\mathrm{L}^{(i)}$
2501.06662_F00088	006	—	$j>i$	$j>i$
2501.06662_F00089	006	—	L_x	L_x
2501.06662_F00090	006	—	$y \in \mathrm{L}$	$y \in \mathrm{L}$
2501.06662_F00091	006	—	$\operatorname{ob}\left(\mathrm{L}_x\right)=\bigsqcup_{j=0}^{N- x } \mathrm{L}_x^{\sim(j)}$	$\operatorname{ob}\left(\mathrm{L}_x\right)=\bigsqcup_{j=0}^{N- x } \mathrm{L}_x^{(j)}$
2501.06662_F00092	006	—	$\mathrm{L}_x^{\sim(j)}$	$\mathrm{L}_x^{(j)}$
2501.06662_F00093	006	—	$ x +j$	$ x +j$
2501.06662_F00094	006	—	$\mathrm{L}_x^{\sim(0)}=\{x\}$	$\mathrm{L}_x^{(0)}=\{x\}$
2501.06662_F00095	006	—	$\mathrm{L}_x^{\sim(1)}=\left\{x a_1: a_1 \in A \cup\{\dagger\}\right\}$	$\mathrm{L}_x^{(1)}=\left\{x a_1: a_1 \in A \cup\{\dagger\}\right\}$
2501.06662_F00096	006	—	$\mathrm{L}_x^{\sim(2)}=\left\{x a_1 a_2: a_1 \in A, a_2 \in A \cup\{\dagger\}\right\}$	$\mathrm{L}_x^{(2)}=\left\{x a_1 a_2: a_1 \in A, a_2 \in A \cup\{\dagger\}\right\}$
2501.06662_F00097	006	—	$\mathrm{L}_{\perp}^{\sim(j)}=\mathrm{L}^{\sim(j)}$	$\mathrm{L}_{\perp}^{(j)}=\mathrm{L}^{(j)}$
2501.06662_F00098	006	—	$j \geq 0$	$j \geq 0$
2501.06662_F00099		—	$\top$	$\top$
2501.06662_F00100	009	—	$\mathbf{L}$	$\mathbf{L}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value				

Identifier	Page	Conf.	LaTeX source	Rendered
2501.06662_F00101	007	—	p_{x}:=p(-\mid x) : A \cup {\dagger}\rightarrow [0,1]	$p_x := p(- \mid x) : A \cup \{\dagger\} \rightarrow [0,1]$
2501.06662_F00102	007	—	a_1={\dagger}	$a_1 = \dagger$
2501.06662_F00103	007	—	p_{\perp a\, a_1}	$P_{\bot aa_1}$
2501.06662_F00104	007	—	A \cup {\dagger}	$A \cup \{\dagger\}$
2501.06662_F00105	007	—	p_{\perp a\, a_1\, a_2}	$P_{\bot aa_1a_2}$
2501.06662_F00106	007	—	y=x\, a_1 \cdots a_{N- x }	$y = xa_1 \cdots a_{N - x }$
2501.06662_F00107	007	—	N	N
2501.06662_F00108	007	—	a=a(x)\in A \cup {\dagger}	$a = a(x) \in A \cup \{\dagger\}$
2501.06662_F00109	007	—	p_x(a)=1	$p_x(a) = 1$
2501.06662_F00110	007	—	p_x(a^{\prime})=0	$p_x(a') = 0$
2501.06662_F00111	007	—	\left.a^{\prime }\in(A\cup {\dagger })\backslash {a}\right)	$a' \in (A \cup \{\dagger\}) \backslash \{a\})$
2501.06662_F00112	007	—	p_{\perp a_1 \cdots a_n} (-)	$p_{\bot a_1 \cdots a_n} (-)$
2501.06662_F00113	007	—	a_n	a_n
2501.06662_F00114	007	—	n ≥ 1	$n \geq 1$
2501.06662_F00115	007	—	\tilde{A}=A \cup {\dagger}	$\tilde{A} = A \cup \{\dagger\}$
2501.06662_F00116	007	—	P_n(a_n,a_{n+1}):=p_{\perp a_1 \cdots a_n}(a_{n+1})	$P_n(a_n,a_{n+1}) := p_{\bot a_1 \cdots a_n}(a_{n+1})$
2501.06662_F00117	007	—	P_n:\,\tilde{A} \times \tilde{A} \rightarrow [0,1]	$P_n:A\times A\rightarrow[0,1]$
2501.06662_F00118	007	—	\sum_{a^{\prime }\in \tilde{A}}P_n(a,a^{\prime })=1	$\sum_{a'\in\tilde{A}} P_n(a,a') = 1$
2501.06662_F00119	007	—	P_n(\,{\dag},\,{\dag})=1	$P_n(\dagger ,\dagger) = 1$
2501.06662_F00120	007	—	P_n(\,{\dag},\,a)=0	$P_n(\dagger ,a) = 0$
2501.06662_F00121	007	—	a ∈ A	$a \in A$
2501.06662_F00122	007	—	n>2	$n > 2$
2501.06662_F00123	007	—	P_n=P_1	$P_n = P_1$
2501.06662_F00124	007	—	a^{\prime }\in A	$a' \in A$
2501.06662_F00125	007	—	P_n(a,\,{\dag})=0	$P_n(a,\dagger) = 0$
2501.06662_F00126	007	—	\left.P_n\right _{A\times A}	$P_n _{A \times A}$
2501.06662_F00127	007	—	\left(P_n\right)_{n\geq 1}	$(P_n)_{n\geq 1}$
2501.06662_F00128	007	—	q:A\rightarrow [0,1]	$q:A\rightarrow[0,1]$
2501.06662_F00129	007	—	\sum_{a^{\prime }\in A} q(a) \, P(a,a^{\prime })=q(a^{\prime })	$\sum_{a'\in A} q(a) P(a,a') = q(a')$
2501.06662_F00130	007	—	p_{\perp }(a)=q(a)	$p_{\bot }(a) = q(a)$
2501.06662_F00131	008	—	n-1	$n - 1$
2501.06662_F00132	008	—	p_{\perp a_1 \cdots a_i}	$p_{\bot a_1 \cdots a_i}$
2501.06662_F00133	008	—	\perp a_1 \cdots a_i	$\bot \, a_1 \cdots a_i$
2501.06662_F00134	008	—	i<n-2	$i < n - 2$
2501.06662_F00135	008	—	p^{(\theta)}:{\cal A}^*\rightarrow \Delta ({\cal A}),\quad a\mapsto p_{\perp a}^{(\theta)}(-)	$p^{(\theta)}:A^* \rightarrow \Delta(A), \quad a \mapsto p_{\bot a}^{(\theta)}(-)$
2501.06662_F00136	008	—	\Delta({\cal A})	$\Delta(A)$
2501.06662_F00137	008	—	{\theta}	θ
2501.06662_F00138	008	—	\left(a_1,\ldots,a_{n-1}\right)	$(a_1,\dots,a_{n-1})$
2501.06662_F00139	008	—	\left(a_1,\ldots,a_n\right)	$(a_1,\dots,a_n)$
2501.06662_F00140	008	—	C ⊂ A^*	$C \subset A^*$
2501.06662_F00141	008	—	y=x\, a^{\prime }	$y = xa'$
2501.06662_F00142	008	—	\left a^{\prime }\right =N-\left x\right	$ a' = N - x $
2501.06662_F00143	008	—	a\, a^{\prime }	aa'
2501.06662_F00144	008	—	y=x\, a^{\prime \prime }{\dag}	$y = xa''\dagger$
2501.06662_F00145	008	—	a^{\prime \prime }\in A^*	$a'' \in A^*$
2501.06662_F00146	008	—	\left a^{\prime \prime }\right \leq N-\left x\right -1	$ a'' \leq N - x - 1$
2501.06662_F00147	008	—	a\, a^{\prime \prime }	aa''
2501.06662_F00148	008	—	N>5	$N > 5$
2501.06662_F00149	008	—	b=	$b =$
2501.06662_F00150	008	—	a_1\, a_2\, a_3\, a_4\, a_5\in A^*	$a_1a_2a_3a_4a_5 \in A^*$
2501.06662_F00151	008	—	a=a_1\, a_2	$a = a_1a_2$
2501.06662_F00152		—	f(x)	$f(x)$
2501.06662_F00153		—	f(x) ≤ N − 1	$ f(x) \leq N - 1$
2501.06662_F00154		—	p_f(x)	$pf(x)$
2501.06662_F00155	009	—	x=\perp a_1 \cdots a_t	$x = \bot \, a_1 \cdots a_t$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value				

Identifier	Page	Conf.	LaTeX source	Rendered
2501.06662_F00156	009	—	$y=x\ a_{t+1}\ \cdots\ a_{t+k}$	$y = xa_{t+1} \cdots a_{t+k}$
2501.06662_F00157	009	—	$k\ \geq\ 1$	$k \geq 1$
2501.06662_F00158	009	—	$\left(a_{i}\right)_{i=1}^{t+k-1}\ \subset\ A$	$(a_i)_{i=1}^{t+k-1} \subset A$
2501.06662_F00159	009	—	$a_{t+k}\ \in\ A\ \cup\ \{\dagger\}$	$a_{t+k} \in A \cup \{\dagger\}$
2501.06662_F00160	009	—	$y_{<t+i}$	$y_{<t+i}$
2501.06662_F00161	009	—	$\perp\ a_1\ \cdots\ a_{t+i-1}$	$\perp\ a_1 \cdots a_{t+i-1}$
2501.06662_F00162	009	—	$y_{<t+1}=x$	$y_{<t+1} = x$
2501.06662_F00163	009	—	$m=0$	$m = 0$
2501.06662_F00164	009	—	$T(x)=\{x\}$	$T(x) = \{x\}$
2501.06662_F00165	009	—	$m=1$	$m = 1$
2501.06662_F00166	009	—	$T(x)=\{x\ a\ \mid\ a\ \in\ A\ \cup\ \{\dagger\}\}$	$T(x) = \{xa\ \ a \in A \cup \{\dagger\}\}$
2501.06662_F00167	009	—	$m\ \geq\ 1$	$m \geq 1$
2501.06662_F00168	009	—	$a\ \in\ A^m$	$a \in A^m$
2501.06662_F00169	009	—	$\left(a^{\prime}\right)_{\prime},\ a^{\left(\prime\prime\right)}\ \in\ A^{m-1}\ \times\ A$	$(a',a'') \in A^{m-1} \times A$
2501.06662_F00170	009	—	$i=m-1$	$i = m - 1$
2501.06662_F00171	009	—	$\pi\left(x\ a^{\prime}\ \mid\ x\right) p\left(\dagger\ \mid\ x\ a^{\prime}\right)$	$\pi\left(xa' \mid x\right) p\left(\dagger \mid xa'\right)$
2501.06662_F00172	009	—	$a^{\prime}\ \in\ A^{m-1}$	$a' \in A^{m-1}$
2501.06662_F00173		—	$\square$	$\square$
2501.06662_F00174	010	—	$(\mathcal{V},\leq,\otimes,1)$	$(\mathcal{V},\leq,\otimes,1)$
2501.06662_F00175	010	—	$(\mathcal{V},\leq)$	$(\mathcal{V},\leq)$
2501.06662_F00176	010	—	$(\mathcal{V},\otimes,1)$	$(\mathcal{V},\otimes,1)$
2501.06662_F00177	010	—	$x\ \otimes\ y\ \leq\ x^{\prime}\ \otimes\ y^{\prime}$	$x \otimes y \leq x' \otimes y'$
2501.06662_F00178	010	—	$x\ \leq\ x^{\prime}$	$x \leq x'$
2501.06662_F00179	010	—	$y\ \leq\ y^{\prime}$	$y \leq y'$
2501.06662_F00180	010	—	$([0,1],\ \leq,\ \cdot,\ 1)$	$([0,1],\leq,\cdot,1)$
2501.06662_F00181	010	—	$a\ b:=a\ \cdot\ b$	$ab := a \cdot b$
2501.06662_F00182	010	—	$a,\ b\ \in\ [0,1]$	$a,b \in [0,1]$
2501.06662_F00183	010	—	$[0,\ \infty],\ \geq,\ +,\ 0$	$[0,\infty],\geq,+,0$
2501.06662_F00184	010	—	$a+\infty:=\infty$	$a + \infty := \infty$
2501.06662_F00185	010	—	$\infty+a:=\infty$	$\infty + a := \infty$
2501.06662_F00186	010	—	$a\ \in\ [0,\ \infty]$	$a \in [0,\infty]$
2501.06662_F00187	010	—	$\mathcal{V}$	$\mathcal{V}$
2501.06662_F00188	010	—	$\mathcal{C}$	$\mathcal{C}$
2501.06662_F00189	010	—	$\operatorname{ob}(\mathcal{C})$	$\mathbf{ob}(\mathcal{C})$
2501.06662_F00190	010	—	$\mathcal{C}(x,\ y)$	$\mathcal{C}(x,y)$
2501.06662_F00191	010	—	$x,\ y,\ z\ \in\ \operatorname{ob}(\mathcal{C})$	$x,y,z \in \mathbf{ob}(\mathcal{C})$
2501.06662_F00192	010	—	$\pi(y\ \mid\ x)=0$	$\pi(y \mid x) = 0$
2501.06662_F00193	010	—	$x=y$	$x = y$
2501.06662_F00194	010	—	$\mathcal{L}(x,\ y):=\pi(y\ \mid\ x)$	$\mathcal{L}(x,y) := \pi(y \mid x)$
2501.06662_F00195	011	—	$x,\ y,\ z$	x,y,z
2501.06662_F00196	011	—	$x\ \rightarrowtail\ y\ \rightarrowtail\ z$	$x \rightarrowtail y \rightarrowtail z$
2501.06662_F00197	011	—	$x\ \neq\ y$	$x \neq y$
2501.06662_F00198	011	—	$y\ \neq\ z$	$y \neq z$
2501.06662_F00199	011	—	$\left(a_i\right)_{i=1}^{t+k+k^{\prime}-1}\ \subset\ A$	$(a_i)_{i=1}^{t+k+k'-1} \subset A$
2501.06662_F00200	011	—	$a_{t+k+k^{\prime}}\ \in\ A\ \cup\ \{\dagger\}$	$a_{t+k+k'} \in A \cup \{\dagger\}$
2501.06662_F00201	011	—	$\pi(y\ \mid\ x)=1$	$\pi(y \mid x) = 1$
2501.06662_F00202	011	—	$\pi(z\ \mid\ y)=\pi(z\ \mid\ x)$	$\pi(z \mid y) = \pi(z \mid x)$
2501.06662_F00203	011	—	$z=y$	$z = y$
2501.06662_F00204	011	—	z	z
2501.06662_F00205	011	—	$\pi(y\ \mid\ x)\pi(z\ \mid\ y)\leq\pi(z\ \mid\ x)$	$\pi(y \mid x)\pi(z \mid y) \leq \pi(z \mid x)$
2501.06662_F00206	011	—	$\pi(y\ \mid\ x),\ \pi(z\ \mid\ x)\ \neq\ 0$	$\pi(y \mid x),\pi(z \mid x) \neq 0$
2501.06662_F00207	011	—	$\pi(z\ \mid\ y)=0,\{ \ }^3$	$\pi(z \mid y) = 0,3$
2501.06662_F00208	011	—	$\pi(z\ \mid\ y),\ \pi(z\ \mid\ x)\ \neq\ 0$	$\pi(z \mid y),\pi(z \mid x) \neq 0$
2501.06662_F00209	011	—	$\pi(y\ \mid\ x)=0.\{ \ }^4$	$\pi(y \mid x) = 0.4$
2501.06662_F00210	011	—	$[0,1]$	$[0,1]$
2501.06662_F00211	011	—	$\mathcal{L}(x,\ y)$	$\mathcal{L}(x,y)$
2501.06662_F00212		—	$x=\perp$	$x = \perp$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value				

Identifier	Page	Conf.	LaTeX source	Rendered
2501.06662_F00213		—	y=\perp	$y = \perp$
2501.06662_F00214		—	z=\perp	$z = \perp$
2501.06662_F00215	012	—	d(x, y)+d(y, z) \geq d(x, z)	$d(x,y) + d(y,z) \geq d(x,z)$
2501.06662_F00216	012	—	d(x, x)=0	$d(x,x) = 0$
2501.06662_F00217	012	—	\infty	∞
2501.06662_F00218	012	—	\mathcal{M}(x, y):=d(x, y)	$\mathcal{M}(x,y) := d(x,y)$
2501.06662_F00219	012	—	\mathrm{L}^{\leq N}	$\mathbf{L}^{\leq N}$
2501.06662_F00220	012	—	R	R
2501.06662_F00221	012	—	\ -\: \operatorname{ob}(\mathcal{V}) \rightarrow R	$\ - \ : \mathbf{ob}(\mathcal{V}) \rightarrow R$
2501.06662_F00222	012	—	\ v\ =\ w\	$\ v\ = \ w\ $
2501.06662_F00223	012	—	v \cong w	$v \cong w$
2501.06662_F00224	012	—	\ 1\ =1	$\ 1\ = 1$
2501.06662_F00225	012	—	\ v \otimes w\ =\ v\ \ w\	$\ v \otimes w\ = \ v\ \ w\ $
2501.06662_F00226	012	—	v, w \in \operatorname{ob}(\mathcal{V})	$v, w \in \mathbf{ob}(\mathcal{V})$
2501.06662_F00227	012	—	\zeta_{\mathcal{C}}\: \operatorname{ob}(\mathcal{C}) \times \operatorname{ob}(\mathcal{C}) \rightarrow R	$\zeta_{\mathcal{C}} : \mathbf{ob}(\mathcal{C}) \times \mathbf{ob}(\mathcal{C}) \rightarrow R$
2501.06662_F00228	012	—	\zeta_{\mathcal{C}}(x, y):=\ \mathcal{C}(x, y)\	$\zeta_{\mathcal{C}}(x,y) := \ \mathcal{C}(x,y)\ $
2501.06662_F00229	012	—	\zeta_{\mathcal{C}}	$\zeta_{\mathcal{C}}$
2501.06662_F00230	012	—	\mu_{\mathcal{C}}:=\zeta_{\mathcal{C}}^{-1}	$\mu_{\mathcal{C}} := \zeta_{\mathcal{C}}^{-1}$
2501.06662_F00231	012	—	\operatorname{Mag}(\mathcal{C})	$\mathbf{Mag}(\mathcal{C})$
2501.06662_F00232	013	—	\mathcal{V}=[0, \infty]	$\mathcal{V} = [0, \infty]$
2501.06662_F00233	013	—	\ x\ _t=e^{-tx}	$\ x\ _t = e^{-tx}$
2501.06662_F00234	013	—	t \in \mathbb{R} \cup \{\infty\}	$t \in \mathbb{R} \cup \{\infty\}$
2501.06662_F00235	013	—	\ -_1	$\ - \ _1$
2501.06662_F00236	013	—	f(t):=\operatorname{Mag}(t \mathcal{C})	$f(t) := \mathbf{Mag}(t\mathcal{C})$
2501.06662_F00237	013	—	t \in (0, \infty)	$t \in (0, \infty)$
2501.06662_F00238	013	—	t \mathcal{C}	$t\mathcal{C}$
2501.06662_F00239	013	—	[s, t]=\{u \in P: s \leq u \leq t\}	$[s,t] = \{u \in P : s \leq u \leq t\}$
2501.06662_F00240	013	—	s \leq t	$s \leq t$
2501.06662_F00241	013	—	\mathbb{k}	$\mathbb{k}$
2501.06662_F00242	013	—	\operatorname{int}(P)	$\mathbf{int}(P)$
2501.06662_F00243	013	—	I(P, \mathbb{k})	$I(P, \mathbb{k})$
2501.06662_F00244	013	—	f: \operatorname{int}(P) \rightarrow \mathbb{k}	$f : \mathbf{int}(P) \rightarrow \mathbb{k}$
2501.06662_F00245	013	—	f(s, t)	$f(s,t)$
2501.06662_F00246	013	—	f([s, t])	$f([s,t])$
2501.06662_F00247	013	—	\delta\: \operatorname{int}(P) \rightarrow \mathbb{k}	$\delta : \mathbf{int}(P) \rightarrow \mathbb{k}$
2501.06662_F00248	013	—	f * \delta=\delta * f=f	$f * \delta = \delta * f = f$
2501.06662_F00249	013	—	f \in I(P, \mathbb{k})	$f \in I(P, \mathbb{k})$
2501.06662_F00250	013	—	\zeta_P	ζ_P
2501.06662_F00251	013	—	\mathbf{2}:=\{	$\mathbf{2} := \{$
2501.06662_F00252	013	—	\}	$\}$
2501.06662_F00253	013	—	s, u \in P	$s, u \in P$
2501.06662_F00254	013	—	s \leq u	$s \leq u$
2501.06662_F00255	013	—	P(s, u)	$P(s,u)$
2501.06662_F00256	013	—	s \not \leq u	$s \not\leq u$
2501.06662_F00257	013	—	\ -\: \operatorname{ob}(\mathbf{2}) \rightarrow \mathbb{Z}	$\ - \ : \mathbf{ob}(\mathbf{2}) \rightarrow \mathbb{Z}$
2501.06662_F00258	013	—	\	$\ $
2501.06662_F00259	013	—	\ =1	$\ = 1$
2501.06662_F00260	013	—	\ =0	$\ = 0$
2501.06662_F00261	013	—	\zeta_P(s, u)=\ P(s, u)\	$\zeta_P(s,u) = \ P(s,u)\ $
2501.06662_F00262	013	—	\mu_P	μ_P
2501.06662_F00263	014	—	\mu_{\mathrm{L}}	$\mu_{\mathbf{L}}$
2501.06662_F00264	014	—	x \in \operatorname{ob}(\mathrm{L})	$x \in \mathbf{ob}(\mathbf{L})$
2501.06662_F00265	014	—	a_1, a_2, \ldots \in A \cup \{\dagger\}	$a_1, a_2, \dots \in A \cup \{\dagger\}$
2501.06662_F00266	014	—	\mu_{\mathrm{L}}(x, x_{a_1} a_{a_2} \cdots a_{a_j})=0	$\mu_{\mathbf{L}}(x, xa_1a_2 \cdots a_j) = 0$
2501.06662_F00267	014	—	j>1	$j > 1$
2501.06662_F00268	014	—	\# A	$\#A$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value				

Identifier	Page	Conf.	LaTeX source	Rendered
2501.06662_F00269	014	—	\#	#
2501.06662_F00270	014	—	(\mathrm{L})	(L)
2501.06662_F00271	015	—	$\mathrm{L}^{\wedge\{\leq N\}}=\mathrm{bigsqcup}_{i=0}^{N-1}\mathrm{L}^{\wedge\{i\}}$	$\mathbf{L}^{\leq N}=\bigsqcup_{i=0}^{N-1}\mathbf{L}^{(i)}$
2501.06662_F00272	015	—	$\mathrm{L}^{\wedge\{(\emptyset)\}}=\{\perp\}$	$\mathbf{L}^{(0)}=\{\perp\}$
2501.06662_F00273	015	—	$i\leq 1,\mathrm{L}^{\wedge\{i\}}$	$i\leq 1,\mathbf{L}^{(i)}$
2501.06662_F00274	015	—	$a\in A^i$	$a\in A^i$
2501.06662_F00275	015	—	$\perp a^{\prime\prime}\dagger$	$\perp a'\dagger$
2501.06662_F00276	015	—	$a^{\prime\prime}\in A^{i-1}$	$a'\in A^{i-1}$
2501.06662_F00277	015	—	$\#\mathrm{L}^{\wedge\{i\}}=\#A^i+\#A^{i-1}$	$\#\mathbf{L}^{(i)}=\#A^i+\#A^{i-1}$
2501.06662_F00278	015	—	$\operatorname{Mag}(\mathrm{L})=1$	$\mathbf{Mag}(\mathbf{L})=1$
2501.06662_F00279	015	—	$\operatorname{ob}(\mathrm{L})=\operatorname{ob}(\mathrm{L}^{\leq N})$	$\mathbf{ob}(\mathcal{M})=\mathbf{ob}\left(\mathbf{L}^{\leq N}\right)$
2501.06662_F00280	015	—	$\mathcal{M}(x,y)=d(x,y)=-\ln \pi(y\mid x)$	$\mathcal{M}(x,y)=d(x,y)=-\ln \pi(y\mid x)$
2501.06662_F00281	015	—	$\zeta_{\mathcal{M}}:\operatorname{ob}(\mathcal{M})\times\operatorname{ob}(\mathcal{M})\rightarrow\mathbb{R}$	$\zeta_{\mathcal{M}}:\mathbf{ob}(\mathcal{M})\times\mathbf{ob}(\mathcal{M})\rightarrow\mathbb{R}$
2501.06662_F00282	015	—	$d(x,y)=\infty$	$d(x,y)=\infty$
2501.06662_F00283	015	—	ζ_t	ζ_t
2501.06662_F00284	015	—	$(\zeta_{\mathcal{M}})_t$	$(\zeta_{\mathcal{M}})_t$
2501.06662_F00285	015	—	ζ_t^{-1}	ζ_t^{-1}
2501.06662_F00286	015	—	$\operatorname{ob}(\mathcal{M})$	$\mathbf{ob}(\mathcal{M})$
2501.06662_F00287	015	—	$x=y_0\rightarrow y_1\rightarrow\cdots\rightarrow y_k=y$	$x=y_0\rightarrow y_1\rightarrow\cdots\rightarrow y_k=y$
2501.06662_F00288	016	—	δ	δ
2501.06662_F00289	016	—	$\delta(x,y)=1$	$\delta(x,y)=1$
2501.06662_F00290	016	—	$\delta(x,y)=0$	$\delta(x,y)=0$
2501.06662_F00291	016	—	$\zeta-\delta$	$\zeta-\delta$
2501.06662_F00292	016	—	D	D
2501.06662_F00293	016	—	$E=\{(z,z')\mid z\rightarrow z'\}$	$E=\{(z,z')\mid z\rightarrow z'\}$
2501.06662_F00294	016	—	$\mathrm{L},z\neq z'$	$\mathbf{L},z\neq z'$
2501.06662_F00295	016	—	$(\zeta_t-\delta)^k$	$(\zeta_t-\delta)^k$
2501.06662_F00296	016	—	k	k
2501.06662_F00297	016	—	E	E
2501.06662_F00298	016	—	(x,y)	(x,y)
2501.06662_F00299	016	—	$\zeta_{\mathbf{L}}^{-1}$	$\zeta_{\mathbf{L}}^{-1}$
2501.06662_F00300	016	—	$\zeta_{\mathbf{L}}$	$\zeta_{\mathbf{L}}$
2501.06662_F00301	017	—	$k\geq 0$	$k\geq 0$
2501.06662_F00302	017	—	$\zeta_t^{-1}(x,y)$	$\zeta_t^{-1}(x,y)$
2501.06662_F00303	017	—	$\pi(y\mid x)^t\zeta_{\mathbf{L}}^{-1}(x,y)$	$\pi(y\mid x)^t\zeta_{\mathbf{L}}^{-1}(x,y)$
2501.06662_F00304	017	—	$\zeta_{\mathbf{L}}^{-1}(x,y)=\mu_{\mathbf{L}}(x,y)$	$\zeta_{\mathbf{L}}^{-1}(x,y)=\mu_{\mathbf{L}}(x,y)$
2501.06662_F00305	017	—	$f:(0,\infty)\rightarrow\mathbb{R}$	$f:(0,\infty)\rightarrow\mathbb{R}$
2501.06662_F00306	017	—	$f(t)=\mathbf{Mag}(t\mathcal{M})$	$f(t)=\mathbf{Mag}(t\mathcal{M})$
2501.06662_F00307	017	—	$p=(p_1,\ldots,p_n)$	$p=(p_1,\ldots,p_n)$
2501.06662_F00308	017	—	$(p_x:A\cup\{\dagger\}\rightarrow[0,1])_{x\in\operatorname{ob}(\mathbf{L})}$	$(p_x:A\cup\{\dagger\}\rightarrow[0,1])_{x\in\mathbf{ob}(\mathbf{L})}$
2501.06662_F00309	017	—	$T(\perp)$	$T(\perp)$
2501.06662_F00310		—	$\operatorname{Mag}(t\mathcal{M})=\sum_{x,y\in\operatorname{ob}(\mathcal{M})}\pi(y\mid x)^t\zeta_{\mathbf{L}}^{-1}(x,y)$	$\mathbf{Mag}(t\mathcal{M})=\sum_{x,y\in\mathbf{ob}(\mathcal{M})}\pi(y\mid x)^t\zeta_{\mathbf{L}}^{-1}(x,y)$
2501.06662_F00311	018	—	$y\neq x$	$y\neq x$
2501.06662_F00312	018	—	$\operatorname{ob}(\mathcal{M})\backslash T(\perp)$	$\mathbf{ob}(\mathcal{M})\backslash T(\perp)$
2501.06662_F00313	018	—	$\operatorname{ob}(\mathcal{M})=T(\perp)\sqcup(\operatorname{ob}(\mathcal{M})\backslash T(\perp))$	$\mathbf{ob}(\mathcal{M})=T(\perp)\sqcup(\mathbf{ob}(\mathcal{M})\backslash T(\perp))$
2501.06662_F00314	018	—	$\zeta_t^{-1}(x,x)=1$	$\zeta_t^{-1}(x,x)=1$
2501.06662_F00315	018	—	$x\in\operatorname{ob}(\mathcal{M})$	$x\in\mathbf{ob}(\mathcal{M})$
2501.06662_F00316	018	—	$t\neq 1$	$t\neq 1$
2501.06662_F00317	018	—	$(t-1)/(t-1)$	$(t-1)/(t-1)$
2501.06662_F00318	018	—	$t\geq 1$	$t\geq 1$
2501.06662_F00319	018	—	$p:S\rightarrow[0,1]$	$p:S\rightarrow[0,1]$
2501.06662_F00320	018	—	S	S
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value				

Identifier	Page	Conf.	LaTeX source	Rendered
2501.06662_F00321	018	—	H(p)	$H(p)$
2501.06662_F00322	018	—	p	p
2501.06662_F00323	019	—	E_{x}(a)=	$E_x(a) =$
2501.06662_F00324	019	—	-\ln p_x(a)=d(x, x a)	$-\ln p_x(a) = d(x, xa)$
2501.06662_F00325	019	—	f(t)=	$f(t) =$
2501.06662_F00326	019	—	\#(\operatorname{ob}(\mathcal{M}))-\tilde{Z}(t)	$\#(\mathbf{ob}(\mathcal{M})) - \tilde{Z}(t)$
2501.06662_F00327	019	—	\tilde{Z}	$\tilde{Z}$
2501.06662_F00328	019	—	(x, a) \in	$(x, a) \in$
2501.06662_F00329	019	—	S:=(\operatorname{ob}(\mathcal{M}) \backslash T(\perp)) \times (A \cup \{\dagger\})	$S := (\mathbf{ob}(\mathcal{M}) \backslash T(\perp)) \times (A \cup \{\dagger\})$
2501.06662_F00330	019	—	E(x, a)=d(x, a)=-\ln p_x(a)	$E(x, a) = d(x, a) = -\ln p_x(a)$
2501.06662_F00331	019	—	\rho	ρ
2501.06662_F00332	019	—	\rho(s)=	$\rho(s) =$
2501.06662_F00333	019	—	e^{-t E(s)} / \tilde{Z}(t)	$e^{-tE(s)} / \tilde{Z}(t)$
2501.06662_F00334	019	—	y, z \in \operatorname{ob}\!\left(\mathcal{M}_x\right)	$y, z \in \mathbf{ob}(\mathcal{M}_x)$
2501.06662_F00335	019	—	\mathcal{M}_x(y,z):=\mathcal{M}(y,z)=-\ln \pi(z \mid y)	$\mathcal{M}_x(y, z) := \mathcal{M}(y, z) = -\ln \pi(z \mid y)$
2501.06662_F00336	020	—	M, d	M, d
2501.06662_F00337	020	—	\ell \in [0, \infty)	$\ell \in [0, \infty)$
2501.06662_F00338	020	—	\left(M\,C_{\{k,\,\ell\}}(M)\right)_{k\,\in\,\mathbb{N}}	$(MC_{k,\ell}(M))_{k \in \mathbb{N}}$
2501.06662_F00339	020	—	M\,C_{\{k,\,\ell\}}(M):=\mathbb{Z}\left[G_{\{k,\,\ell\}}\right]	$MC_{k,\ell}(M) := \mathbb{Z}[G_{k,\ell}]$
2501.06662_F00340	020	—	\partial_k: M\,C_{\{k,\,\ell\}}(M) \rightarrow M\,C_{\{k-1,\,\ell\}}(M)	$\partial_k : MC_{k,\ell}(M) \rightarrow MC_{k-1,\ell}(M)$
2501.06662_F00341	020	—	0< i < k	$0 < i < k$
2501.06662_F00342	020	—	\partial_k^0=\partial_k^k=0	$\partial_k^0 = \partial_k^k = 0$
2501.06662_F00343	020	—	[0, \infty)	$[0, \infty)$
2501.06662_F00344	020	—	\left(M\,C_{\{\bullet,\,\ell\}}(M),\,\partial_{\bullet}\right)	$(MC_{\bullet,\ell}(M), \partial_{\bullet})$
2501.06662_F00345	020	—	M	M
2501.06662_F00346	020	—	H_{\{\bullet,\bullet\}}(M)	$H_{\bullet\bullet}(M)$
2501.06662_F00347	020	—	\left(M\,C_{\{\bullet,\ell\}},\,\ell_{\ell}\right)_{\ell}	$(MC_{\bullet,\ell})_{\ell}$
2501.06662_F00348	021	—	q=e^{-t}	$q = e^{-t}$
2501.06662_F00349	021	—	\bigcup_{k\geq 0} M\,C_{\{k,\,\ell\}}(\mathcal{M}) \neq	$\bigcup_{k \geq 0} MC_{k,\ell}(\mathcal{M}) \neq$
2501.06662_F00350	021	—	\emptyset	$\emptyset$
2501.06662_F00351	021	—	x=y_{\{0\}} \rightarrow	$x = y_0 \rightarrow$
2501.06662_F00352	021	—	y_{\{1\}} \rightarrow \cdots \rightarrow y_k = y	$y_1 \rightarrow \cdots \rightarrow y_k = y$
2501.06662_F00353	021	—	\prod_{i=1}^k \pi\!\left(y_{\{i\}} \mid y_{\{i-1\}}\right) \neq 0	$\prod_{i=1}^k \pi(y_i \mid y_{i-1}) \neq 0$
2501.06662_F00354	021	—	d\!\left(y_{\{i\}},\,y_{\{i+1\}}\right)<\infty	$d(y_i, y_{i+1}) < \infty$
2501.06662_F00355	021	—	i=0,\,\ldots,\,k-1	$i = 0, \dots, k-1$
2501.06662_F00356	021	—	\ell:=\sum_{i=0}^{k-1} d\!\left(y_{\{i\}},\,y_{\{i+1\}}\right)<\infty	$\ell := \sum_{i=0}^{k-1} d(y_i, y_{i+1}) < \infty$
2501.06662_F00357	021	—	\left(y_{\{0\}},\,\ldots,\,y_{\{k\}}\right)	$(y_0, \dots, y_k)$
2501.06662_F00358	021	—	G_{\{k,\,\ell\}}	$G_{k,\ell}$
2501.06662_F00359	021	—	\prod_{i=1}^k \pi\!\left(y_{\{i\}} \mid y_{\{i-1\}}\right)^t=\exp\left(-t\,\ell\right)	$\prod_{i=1}^k \pi(y_i \mid y_{i-1})^t = \exp(-t\ell)$
2501.06662_F00360	021	—	M\,C_{\{k,\,\ell\}}	$MC_{k,\ell}$
2501.06662_F00361	021	—	\left(M\,C_{\{\bullet,\ell\}}\right)_{\ell}	$(MC_{\bullet,\ell})_{\ell}$
2501.06662_F00362	022	—	H_{\{0,0\}}(\mathcal{M})	$H_{0,0}(\mathcal{M})$
2501.06662_F00363	022	—	\ell>0	$\ell > 0$
2501.06662_F00364	022	—	H_{\{0,\,\ell\}}(\mathcal{M})=0	$H_{0,\ell}(\mathcal{M}) = 0$
2501.06662_F00365	022	—	H_{\{1,\,\ell\}}(\mathcal{M})	$H_{1,\ell}(\mathcal{M})$
2501.06662_F00366	022	—	d(x, y)=\ell	$d(x, y) = \ell$
2501.06662_F00367	022	—	y=a_{\{0\}}\,a_{\{1\}}\,\cdots\,a_{\{n\}}	$y = a_0 a_1 \cdots a_n$
2501.06662_F00368	022	—	p\!\left(- \mid y_{<i}\right)	$p(- \mid y_{<i})$
2501.06662_F00369	022	—	y_{<i}	$y_{<i}$
2501.06662_F00370	022	—	\zeta_t\!\left(a_{\{0\}},\,y\right)	$\zeta_t(a_0, y)$
2501.06662_F00371	022	—	t=1 \,/\, n	$t = 1/n$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value				

Identifier	Page	Conf.	LaTeX source	Rendered
2501.06662_F00372	022	—	$a_{\{0\}}$	a_0
2501.06662_F00373	022	—	$\zeta_t\!\left(a_{\{0\}},\,y\right)=1\,/\,P\,P\,L(y)$	$\zeta_t(a_0,y)=1/PPL(y)$
2501.06662_F00374	022	—	$\zeta_t\!\left(x,\,y\right)$	$\zeta_t(x,y)$
2501.06662_F00375	022	—	$\widehat{\mathcal{M}}\!:=\!\left[0,\,\infty\right]^{\mathcal{M}}$	$\widehat{\mathcal{M}}:=\left[0,\infty\right]^{\mathcal{M}}$
2501.06662_F00376	022	—	$f\colon\mathcal{M}\rightarrow\left[0,\,\infty\right]$	$f:\mathcal{M}\rightarrow\left[0,\infty\right]$
2501.06662_F00377	022	—	$\max\left\{f(y)-f(x),\,0\right\}\leq\mathcal{M}(x,\,y)$	$\max\{f(y)-f(x),0\}\leq\mathcal{M}(x,y)$
2501.06662_F00378	022	—	$\widehat{\mathcal{M}}$	$\widehat{\mathcal{M}}$
2501.06662_F00379	023	—	$p\colon\operatorname{ob}(\mathrm{A})\rightarrow\left[0,1\right]$	$p:\operatorname{ob}(\mathbf{A})\rightarrow\left[0,1\right]$
2501.06662_F00380	023	—	$\theta\colon\operatorname{ob}(\mathrm{A})\times\operatorname{ob}(\mathrm{A})\rightarrow\left[0,\infty\right)$	$\theta:\operatorname{ob}(\mathbf{A})\times\operatorname{ob}(\mathbf{A})\rightarrow\left[0,\infty\right)$
2501.06662_F00381	023	—	$\theta=\zeta_t$	$\theta=\zeta_t$
2501.06662_F00382	023	—	$p=\left.\pi(-\mid x)\right _{T(x)}$	$p=\left.\pi(-\mid x)\right _{T(x)}$
2501.06662_F00383	023	—	$\left.\pi(-\mid x)\right _{T(x)}$	$\pi(-\mid x) _{T(x)}$
2501.06662_F00384	023	—	$P\colon A\times A\rightarrow\left[0,1\right]$	$P:A\times A\rightarrow\left[0,1\right]$
2501.06662_F00385	023	—	$p_{\{\perp\}}$	$p_{\perp}$
2501.06662_F00386	024	—	$\{ \ }^{\sim}\!\sim$	$\sim$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value				